RDB-DINO: An Improved End-to-End Transformer With Refined De-Noising and Boxes for Small-Scale Ship Detection in SAR Images

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***Abstract*— Recently, convolution neural networks (CNNs) have been extensively utilized in synthetic aperture radar (SAR) ship detection owing to their strong feature extraction and representation capability. However, existing CNN-based SAR ship detectors often suffer from poor sensitivity to small-scale ship targets due to the limited extractable features, especially in complex inshore scenarios. Moreover, the hand-designed components like nonmaximum suppression (NMS) calculation and anchor generation imposed in CNN-based detector significantly affect their robustness. In the face of these challenges, a novel end-to-end (E2E) transformer-based detection framework for small-scale ship targets in SAR images, named detection transformer (DETR) with improved de-noising (DN) anchor box (DINO) with refined DN and box (RDB-DINO), is proposed in this article. First, we introduce a complete contrastive DN (CCD) training technique which reconstructs and exploits different kinds of noised queries to reduce the confusion between small ships and complex backgrounds. Second, a look twice toward maximum (LTTM) algorithm for iterative box refinement is designed to mine the abnormal sample information and obtain abundant features of small ships in the training process. Finally, substantial experiments conducted on two widely used open SAR ship datasets demonstrate that the proposed approach yields superior results in small ship detection performance, outperforming prevailing state-of-the-art (SOTA) benchmarks.**

***Index Terms*— End-to-end (E2E) transformer, small-scale ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor for all-day and all-weather Earth observation tasks [2], [3]. Concurrently, ship detection plays a vital role in marine situational awareness and monitoring tasks, including resource exploitation, traffic transportation, port management, and emergency rescue [4]. Therefore, ship detection in SAR images has emerged as a significant research direction and attracted widespread attention for decades.

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Traditional methods usually rely on statistical models such as constant false alarm rate (CFAR) [5]. These methods have high requirements for the accuracy of background clutter modeling. Commonly used CFAR algorithms include CA-CFAR [6] and OS-CFAR [7]. Besides, Zhang et al. [8] proposed an unsupervised ship detection method based on superpixels and CSPNet, which automatically detects ships by combining complex signal kurtosis and local superpixel contrast. Cheng et al. [9] proposed a detection framework based on saliency enhancement and Bhattacharyya-like distance, which is particularly suitable for detecting ships in near-shore areas. In addition, Xiang et al. [10] proposed an improved sidelobe suppression method to process SAR images and achieve significant accuracy improvement in target detection. Nevertheless, the traditional methods often have subpar detection performance under complex backgrounds.

Driven by the remarkable success of convolution neural networks (CNNs) for object detection [11], CNN-based detectors have been extensively applied in remote sensing image processing, and achieve superior performance compared with the traditional approaches owing to their powerful feature extraction and representation capabilities [12], [13]. Depending on the different generation ways of anchor candidates, existing CNN-based ship detectors can be divided into two categories: anchor-based and anchor-free architectures [14]. The former excels in complex scenarios by adopting predefined boxes to fit targets [15], while the latter facilitates the resolution of multiscale issues through adaptive box generation [16]. For instance, Qin et al. [17] investigated an improved anchor-based method based on self-distillation for SAR inshore detection. Sun et al. [18] exploited an efficient anchor-free framework for multiscale ship target detection based on the electromagnetic scattering property of ships in SAR images. However, the bottlenecks of CNN-based ship detectors persist for small-scale ships (i.e., occupy less than 32×32 pixels labeled by the format of Microsoft Common Objects in Context dataset [19]) since the small-scale ship targets lack of exhaustive and well-learned feature information. In addition, small ships are weak-intensity in SAR images, making them prone to confusion with speckle noise in complex inshore scenarios. To tackle these issues, most CNN-based ship detectors endeavor to exploit contextual information of targets [20]. However, these methods fall short of capturing long-range interactions and dependencies between targets in images limited by the local receptive fields. In addition,

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the hand-designed components (i.e., unlearnable prior anchor knowledge), such as intersection over union (IoU) threshold and nonmaximum suppression (NMS), weaken the robustness of the model.

Following the success of transformer architectures in natural language processing, vision transformer has also witnessed a prominent growth in many tasks, including image classification, change detection, and object detection [21]. Compared with CNNs, a vision transformer yields a self-attention mechanism based on elements calculation in the sequence to model the content-dependent global interactions and better learn the long-range contextual information. Therefore, transformer-based detectors are considered to be alternatives to CNNs in small object detection. As a task-specific application of vision transformer, transformer-based detectors can be divided into two categories [22]. The first is the transformer backbone methods, which encapsulates the transformer structure into a backbone or a module for feature extraction in CNNs [23]. Intuitively, such hybrid architecture leverages the global modeling capability in the transformer and follows the detection pipeline in CNN to combine the merits of both with strong operability. For example, a Swin transformer is proposed in [24] as a novel backbone based on a shifted window mechanism to replace the commonly used ResNet [25] for feature representation. However, this type of network still follows the detection paradigm of CNN which also requires additional setting of unlearnable parameters about prior anchor, leading to moderate robust performance. The second is the end-to-end (E2E) transformer method, which models the task as a set prediction problem to eliminate the hand-crafted components in CNNs [26]. Detection transformer (DETR) [27] is a pioneer of E2E transformer-based detector to completely migrate the encoder–decoder schema in transformer structure by one-to-one Hungarian matching algorithm. Obviously, the E2E approach breaks free from the constraints of CNN with excellent robustness, and does not require hyperparameters setting in the inference stage. As an emerging detection architecture, the performance of E2E transformer-based detectors still cannot compete with existing state-of-the-art (SOTA) CNN-based approaches due to the slow and unstable training process caused by Hungarian matching. Recently, a series of DETR-liked methods attempted to address the training problems caused by cross-attention in decoder, and have been a new research hotspot in object detection. For instance, a denoising (DN) training method is proposed to construct subsets of noise queries with randomly flipped label embeddings and act as an else auxiliary task to bypass the Hungarian match [28]. However, the method in [28] trends to overlook the information of ground truth (GT) query in DN training and the relationship between noised queries and GTs, leading to unsatisfactory performance gains in complex background. To tackle the issues of insufficient target features in box prediction, an improved box refinement algorithm is designed to promote feature fusion among adjacent decoder layers [29]. Yet the inadequate utilization of box information restricts the effect of feature enhancement, especially for small-scale targets. Specifically, box information refers to the specific information of the bounding box generated during the target detection process, including the center coordinates, width, and height of the bounding box. Generally, the samples of the largest box are usually rare samples or those difficult samples that are incorrectly predicted in small target scenarios. By digging deeper into box information, especially their size information, the model can better focus on these key and valuable samples, thereby enhancing the learning of these samples during the training process and improving the performance of the overall algorithm. Therefore, the box information is crucial for the feature enhancement process when the model detects small-scale targets. However, existing E2E transformer methods do not achieve substantial enhancements in the accuracy of small target detection, especially in complex environments.

Through the above observations, a novel E2E transformer-based detection framework for small-scale ship targets in SAR images, named DETR with improved DN anchor box (DINO) [29] with refined DN and box (RDB-DINO), is proposed in this article. In order to alleviate confusion between small ships and the background, a complete contrastive DN (CCD) training strategy is proposed to achieve straightforward and high-quality anchor selection and partially replace the complicated Hungarian match in loss calculation. Specifically, the proposed CCD strategy introduces GT information as positive samples in DN training, and introduces the IoU with GT in the noise query to fix label embedding. In this way, the model can better learn information from samples, thereby more accurately distinguishing ship targets from background. This is particularly important for small target detection, because small targets usually appear as weak signals in SAR images and are easily confused with complex backgrounds. In addition, CCD directly performs loss calculation by constructing multiple query groups containing GT, positive samples, and negative samples, and partially replaces the traditional Hungarian matching algorithm. This method can effectively utilize different types of sample information, allowing the model to fully learn various features during the training process, thereby improving the model's perception of small targets. Hence, the CCD strategy directly performs one-to-one directional prediction and loss calculation by constructing multiple query groups containing labels, positive samples, and negative samples. This method avoids the complex bipartite graph matching process in the Hungarian matching algorithm and simplifies the training process. Since CCD has already clarified the relevance of the target through the constructed query group, it is no longer necessary to match the query to the target through Hungarian matching. Moreover, we design a look twice toward maximum (LTTM) algorithm in box prediction to mine the abnormal sample information and enhance the features of small ships. Different from the feature fusion among adjacent layers, we select the layer with maximum modulus information of output (i.e., the maximum two-norm of the vector containing width and length values of all the output boxes at the current layer) for feature enhancement to realize the integration of maximum box information. Our main contributions can be summarized as follows.

1) We propose the CCD training technique to enrich the variety of noised queries and modify the label embeddings based on IoU between GT and noised query,

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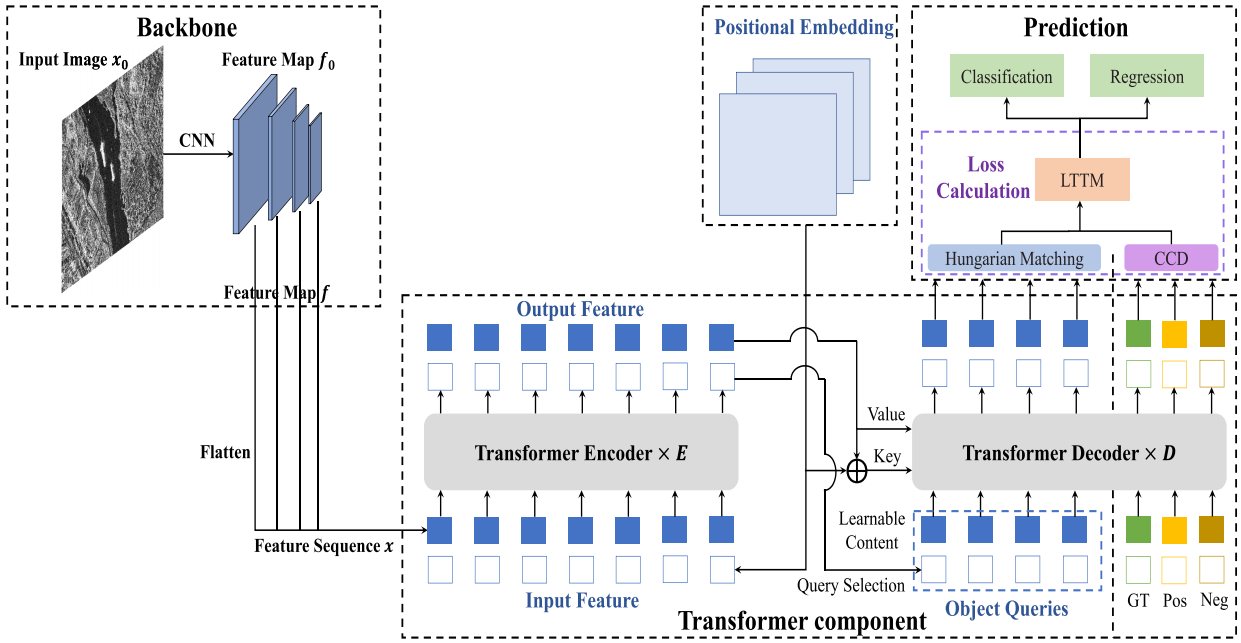


Fig. 1. Overall pipeline of the proposed RDB-DINO. In our architecture, we use DINO [29] as the baseline. The proposed network consists of three parts: backbone, transformer component, and prediction. In the backbone stage, we employ canonical ResNet-101 [25] to extract the feature maps from input images. The transformer component is used for feature enhancement and utilization. Finally, the classification and regression branches output the results based on loss calculation in the prediction head. The GT, Pos, and Neg are short for ground truth, positive sample, and negative sample, respectively. The solid and hollow boxes in the transformer component indicate the content and position features of queries, respectively.

thereby improving the accuracy of small ship detection, especially in inshore environments.

2) We design the LTTM algorithm for iterative box refinement in prediction to achieve sufficient feature mining and enhancement for small ship targets by exploiting the abnormal sample information of the prediction box.

Extensive experiments validate the superiority of our schema in small ship detection over the existing SOTA approaches, especially in inshore scenarios.

The remainder of this article is organized as follows. Section II briefly reviews recent research works in the field of CNN-based small-scale ship detection and transformer-based SAR ship detection. Section III introduces the proposed framework in detail. Experimental verifications of the proposed techniques have been carried out and analyzed in Section IV. Finally, all the conclusions are displayed in Section V.

II. Related Work

In this section, brief descriptions of previous CNN-based small ship detectors and transformer-based SAR ship detection methods are provided. In addition, DETR-like methods are also reviewed.

A. CNN-Based Detectors for Small Ships

Due to the small scale and weak intensity of small ship targets in SAR images, CNN-based detectors mainly adopt the attention mechanism or extension of receptive fields to enhance the salient features and long-range contextual information in small ship detection. In [30], a dense attention pyramid network was proposed to extract feature maps with higher resolution and more semantic information. Wang et al. [31] designed a nonlocal attention mechanism module in a baseline network to enhance the feature extraction ability for small ships by enriching nonlinear relationships among targets with different scales. In FBUA-Net [32], a global context-guided feature balanced pyramid structure integrated united attention module was investigated to construct long-range dependencies among small ship targets. Gong et al. [33] developed a context attention module to make the network pay more attention to small-scale ships in feature extraction. In addition, receptive field amplification is another vital technique for acquiring abundant contextual information. Yue et al. [34] introduced a high-quality anchors generation strategy and receptive field enhancement module to locate and capture the small ship features. Deng et al. [35] redesigned an improved feature extractor by adopting a concatenated inception module to increase the receptive fields of small ships. There are also some works both employ attention mechanism and receptive field expansion to achieve remarkable results for small ship detection. Yang et al. [36] proposed a one-stage ship detector with a coordinate attention module and a receptive field increased module to capture more representative features and contextual information of small-scale ships. However, limited by the local modeling of the convolutional layer, CNN-based detectors still have obstacles and bottlenecks in small-scale ship detection. In addition, the hand-designed procedure imposed in CNN seriously affects the robustness performance of ship detection.

B. Transformer-Based Methods and Their Developments in SAR Ship Detection

Compared with the CNN framework, the vision transformer has the benefit of modeling global dependency with contextual information to tackle the issues in small-scale

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ship detection. In [37], a convolution and transformer fusion network was designed for small-size ship detection. Zhou et al. [38] established a pyramid vision transformer paradigm for small-scale feature representation to achieve satisfactory performance in dense ship distribution. Zha et al. [39] introduced an improved transformer structure to replace the last convolutional layer in the backbone stage and to enrich contextual information for small targets. To differentiate the small ships from the background clutter, a feature transformer module was incorporated into YOLOX [40] to improve global feature extraction capability [41]. Xia et al. [42] proposed an improved backbone combining CNN and transformer to achieve contextually joint representation learning for SAR ship detection. Shi et al. [43] developed a sparse transformer backbone aggregated local information to suppress background clutter, especially for small-scale ships. We can observe that the vast majority of existing transformer-based ship detectors employ the transformer backbones due to the ease of operation and certain performance improvements. However, these detectors still follow the CNN detection pipeline, leading to weak robustness and insignificant improvements in small-ship detection.

As a novel detection paradigm, the E2E transformer-based detector has been the mainstream framework in object detection due to the elimination of necessary hand-designed procedures in CNNs [44]. DETR [27] was the pioneer to achieve the E2E detection without adding hand-designed components like NMS and anchor generalization. The subsequent papers attended to addressing the unstable training and slow convergence produced by decoder cross-attention in the DETR structure. For example, Deformable DETR constructed a deformable attention mechanism which only focused on a few sampling points surrounding a reference point to significantly reduce the computational overhead of the network [45]. Liu et al. [46] explicitly represented the 2-D query point as a 4-D anchor box including center coordinate and size in the decoder layer to alleviate learning difficulty and speed up training. DN-DETR [28] introduced a DN training technique to accelerate the convergence according to bypassing the unstable bipartite matching process. However, these methods were still unable to compete with CNN in terms of accuracy. In pursuit of accuracy, DINO [29] exploited several feasible techniques and acquired the ability to surpass CNN performance in the COCO dataset, sparking interest in the application of the E2E transformer-based approach in SAR ship detection. TSDet [47] and OEGR-DETR [48] explored the application of the E2E transformer-based method within the realm of SAR ship detection, whereas both marginally outperformed the initial DETR and had mediocre performance in small ship detection.

Motivated by the above observations, we design an improved E2E transformer with several feasible techniques in SAR ship detection. Specifically, we proposed an improved DINO-guided detector with a CCD training strategy and LTTM algorithm to achieve inspiring progress on small-scale ship detection, especially in complex inshore scenarios.

III. Methodology

In this section, the proposed RDB-DINO is elaborated. Specifically, the overall pipeline and components of the proposed framework are briefly illustrated in Fig. 1. Then, the proposed components are introduced for small-scale ship detection. Finally, the loss functions are constructed in detail.

A. Overall Pipeline of Our Network

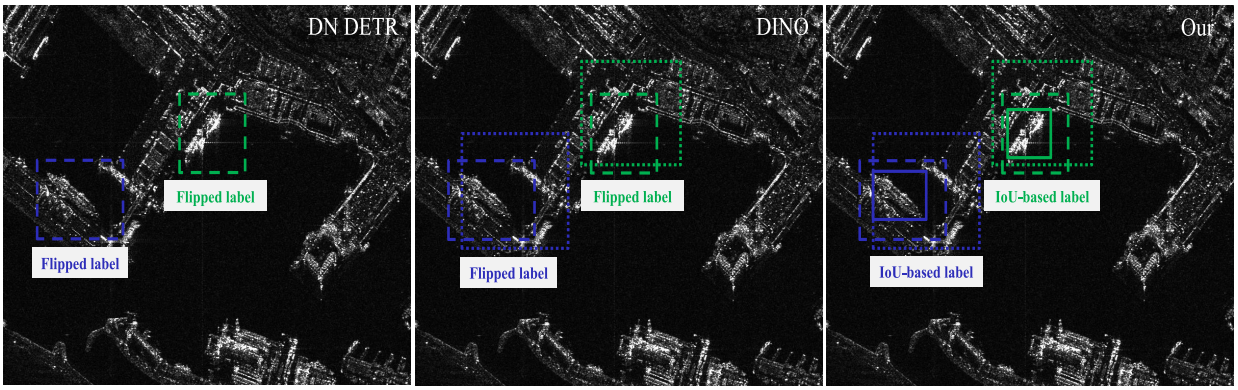
In our proposed RDB-DINO, we employ DINO [29] as the baseline, construct a CCD training strategy in the decoder, and introduce the LTTM algorithm for box refinement in prediction to acquire prominent progress on small-scale ship detection, especially in complex inshore scenarios.

1) Backbone: The SAR image $x_0 \in \mathbb{R}^{3\times H_0\times W_0}$ with $H_0 \times W_0$ pixels and three channels is input into the canonical ResNet-101 [25] network pretrained on ImageNet. It should be noted that we replicate the single channel of SAR images twice to accommodate the model's input. The feature map $f_0 \in \mathbb{R}^{C_0\times H\times W}$ outputs by multiple down-samplings, where $C_0 = 2048$ and $H, W = (H_0/32), (W_0/32)$. Then, a 1×1 convolution is utilized to reduce the number of channels of f_0 and obtain the feature map $f \in \mathbb{R}^{C\times H\times W}$ with $C = 256$ channel dimension. Since the input expected by the encoder in the transformer structure is the formulation of sequence, we collapse the 2-D spatial resolution of f into one dimension to acquire the final pixel feature sequence $x \in \mathbb{R}^{HW\times C}$ by the flatten operation.

2) Transformer Component: As illustrated in Fig. 1, the transformer architecture is composed of two parts, encoder and decoder. In the encoder, the input feature sequence is supplemented with positional embedding through the sine encoding function [27] to locate the features. Hence, we can conceptualize the features in the transformer as a concatenation of content and position. In the decoder, object queries are learnable embeddings with the same format as the features in the encoder. Among them, the content information is a series of learnable numbers initialized from value 0, while the position information is transmitted from encoder output after query selection mechanism [29].

3) Prediction: As presented in Fig. 1, the prediction head contains two subbranches: classification and regression. Similar to DINO [29], the classification branch outputs the class probability after a linear projection layer with a Softmax function, while the regression branch predicts the bounding box including center coordinates, height, and width via the three-layer perceptron with a ReLU activation function. The number of learnable object queries is fixed at $N = 900$, which empirically far exceeds the actual number of ships in each SAR image chip (such as the average of 2.12 ships per image in the public SAR Ship Detection Dataset (SSDD) [33]). Therefore, most of the prediction boxes are judged as no object with label $\varnothing$ (null set) to represent background through the Hungarian matching algorithm. In such a way, the proposed detector can directly predict the bounding boxes and classes (including ship and background) in SAR images to remove the hand-designed components in CNN, thus possessing excellent robustness [27], [47].

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(a) (b) (c)

Fig. 2. Visualization of training strategy in different methods. The green and blue boxes denote the different ship targets. The solid box, dashed box, and dotted dashed box represent the GT, GT with weak noise, and GT with strong noise, respectively. (a) DN training strategy in DN-DETR. (b) CDN training strategy in DINO. (c) CCD training strategy in our model.

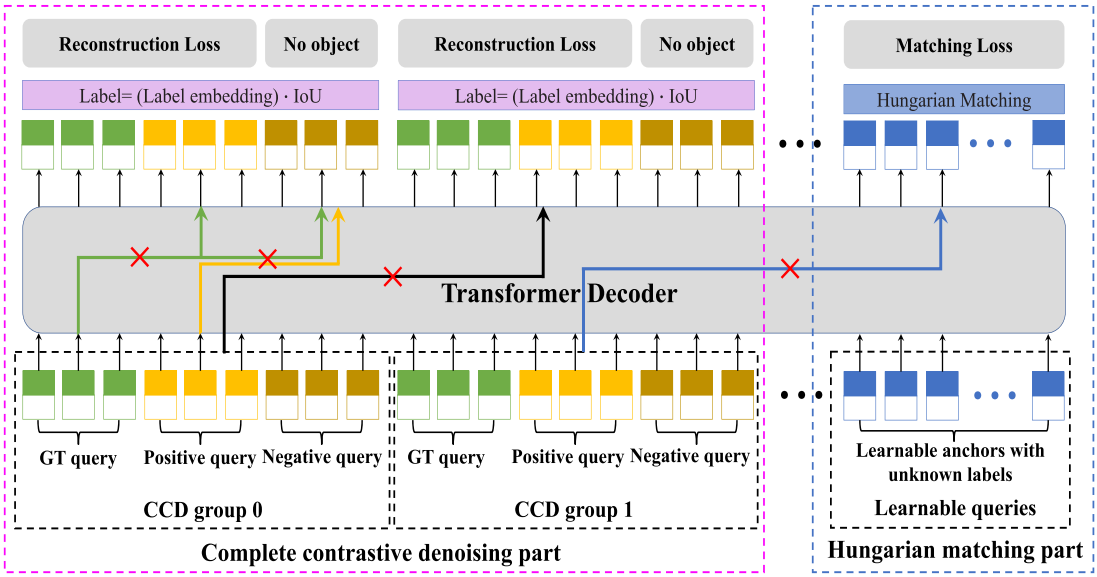


Fig. 3. Overall structure of the proposed CCD strategy. The arrow represents information transfer in the same type of query. The red ✖ mark indicates an attention mask to prevent information leakage among different types of queries. Intuitively, the proposed CCD groups act as additional auxiliary tasks for one-to-one directional prediction and loss calculation to avoid Hungarian matching.

B. Complete Contrastive De-Nosing Training Strategy

As mentioned above, DETR-like detectors employ a Hungarian matching strategy to construct a one-to-one correspondence between queries and targets. However, the matching process may lead to ambiguous and inefficient training. Especially for ship detection in SAR images, the small-scale ships with weak-intensity will exacerbate the matching difficulty. To streamline the initial Hungarian algorithm, a DN training approach is proposed in the E2E transformer detector. As shown in Fig. 2(a), DN-DETR [27] introduces noise queries by adding minor perturbations on GTs as the positive samples to directly predict targets without the matching operation. Benefiting from the DN queries, DN-DETR is effective in accelerating the convergence of the model. However, DN-DETR lacks the capability to withstand interference from redundant useless anchors (queries) near ship targets, especially in complex scenarios. On the basis of DN-DETR, DINO [29] designs a CCD training to discard the useless anchors via additionally introducing queries with strong noise as negative samples which are used to directly predict "no object," as shown in Fig. 2(b). However, these methods trend to overlook the information of GT query in DN training. In addition, the existing de-noise training methods adopt the randomly flipped label embeddings (i.e., randomly flip the GT labels to other categories labels) to label the class probability, causing the network to ignore the association of queries and GTs. To address these issues, we propose a novel CCD training strategy to leverage the information from different query samples and allocate labels properly according to the actual association between the query samples and GTs. As illustrated in Fig. 2(c), we add GTs as supplements to positive samples to directly predict ships in DN groups, aiming to fully utilize the information from various samples. In addition, we use the IoU between noised queries and GTs as the modified label embeddings of the samples to model the relevancy between query and label. The specific architecture of CCD is presented in Fig. 3. In the CCD training process, we construct several

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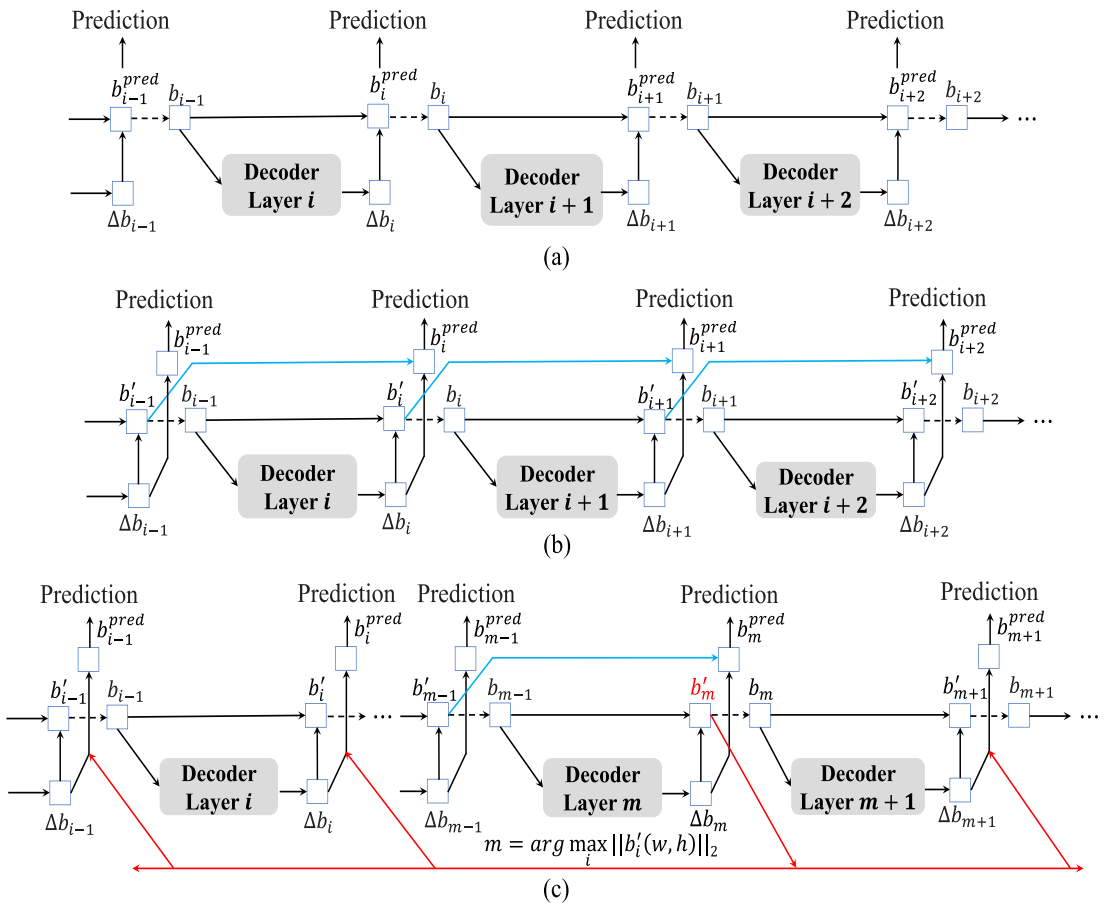


Fig. 4. Box refinement strategies in different models. The dashed arrow indicates gradient truncation. The black, blue, and red arrows denote the gradient propagation in the current decoder layer, from the following decoder layer, and from the decoder layer with maximum modulus information, respectively. (a) Box information update in Deformable DETR. (b) Box information update in DINO. (c) Box information update in our proposed RDB-DINO.

additional sets of GT query, positive query, and negative query as the CCD groups to directly perform loss calculation and partially replace the Hungarian matching. Therefore, the proposed network effectively leverages the information from different queries and achieves preferable training stability. Similar to DINO [29], we utilize two parameters λ_1 and λ_2 to control the ratio of added noises for positive and negative queries, where $\lambda_2 > \lambda_1$.

C. LTTM Algorithm in Box Prediction

In SAR ship detection, small-scale ships are susceptible to be missed due to their small size and few features. In order to promote the model's perception of small targets, we propose the LTTM scheme in box refinement via adding the maximum predicted box information. As shown in Fig. 4(a), the prediction b_i^{pred} is only determined by the output Δb_i of the current decoder layer in Deformable DETR [45]. However, DINO [29] conjectures that abundant box information from the following layer is beneficial to box refinement, thus puts forward a look forward twice (LFT) algorithm to incorporate the box information from the subsequent decoder layer into the current decoder layer for box prediction. Intuitively, the prediction result b_i^{pred} in DINO is depended on $(b'_i, \Delta b_i)$, as shown in Fig. 4(b). Inspired by DINO and hard example mining, we design another way to identify and emphasize anomalous output results during the training process. Our insight is to mine and propagate the anomalously large bounding boxes predicted by the decoders in the training process to other layers, correcting significant erroneous predictions for small objects during training. As presented in Fig. 4(c), we filter out the box information with maximum width w and height h modulus among the outputs (i.e., b'_m) of decoder layers, and then transfer such information to other decoder layers for box prediction, while for the layer with maximum box information, we employ the same method as LFT for box refinement. Compared to the LFT algorithm, our approach achieves the integration of maximum box information with current prediction offset, and significantly enhances the perception of abnormal prediction boxes. Hence, we call our method as "LTTM."

As illustrated in Fig. 4(c), the quality of predicted box b_i^{pred} at the ith decoder layer is influenced by two factors: the box information b'_m at the $m = \arg\max_i ||b'_i(w,h)||_2$ layer with maximum width and height modulus, and the output offset of the box Δb_i at current decoder layer. Hence, the box refinement process can be formulated as

$$b_i^{\text{pred}} = \text{Update}\left(b'_m, \Delta b_i\right) \tag{1}$$

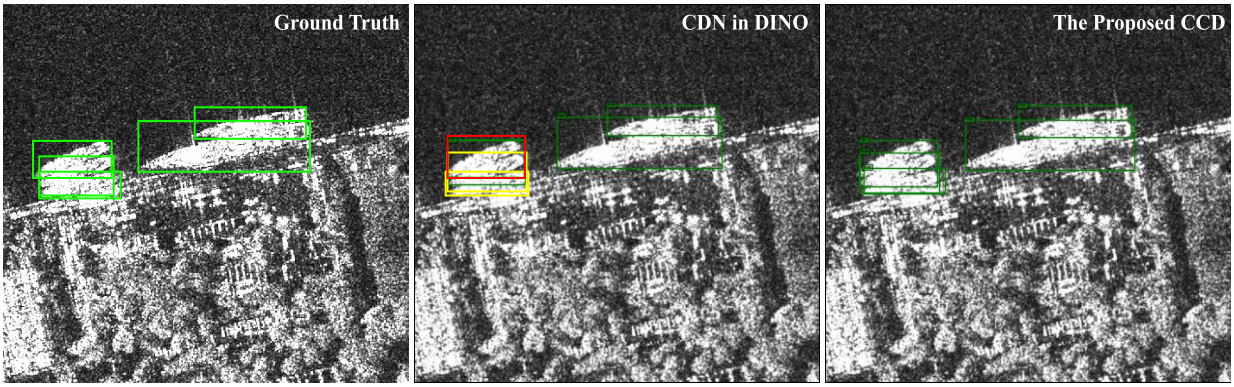
$$\Delta b_i = \text{Layer}_i(b_{i-1}) \tag{2}$$

$$b'_m = \text{Update}(b_{m-1}, \Delta b_i) \tag{3}$$

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TABLE I
DESCRIPTIONS OF EXISTING TWO DATASETS IN SAR SHIP DETECTION

Dataset	Satellite	Polarization	Resolution	Image number	Image size	Ship number	Number of small ships
SSDD	RadarSat-2, TerraSAR-X, Sentinel-1	Full	1-15	1160	500 × 500, etc	2456	1529
HRSID	Sentinel-1B, TerraSAR-X, TanDem	HH, VV, HV	0.5, 1, 3	5604	800 × 800	16965	9242



(a) (b) (c)

Fig. 5. Detection visualization results obtained by different training methods. The green, red, and yellow rectangles indicate the correctly detected ship, false alarms, and missed detection, respectively. (a) GT. (b) Detection results using a model trained with CDN. (c) Detection results using a model trained with the proposed CCD.

TABLE II
PARAMETERS SETTING IN OUR MODEL

Parameter Item	Value
Learning rate	0.0001
Learning rate drop	100
Epoch	150
Encoder number E	6
Decoder number D	6
Transformer attention head M	8
Query number q	900
Feature level L	4
Sampling point K	4
CCD group N_g	100
Positive noise hyperparameter λ_1	0.2
Negative noise hyperparameter λ_2	0.4
GIoU loss coefficient λ_{GIoU}	2.0
L1 loss coefficient λ_{L1}	5.0
Classification loss coefficient λ_f	1.0
Sample balance coefficient α	0.25
Focal loss factor γ	2

$$b_{m-1} = \text{Detach}\left(b'_{m-1}\right) \tag{4}$$

where the terms $\text{Layer}_i(\cdot)$ and $\text{Detach}(\cdot)$ denote the ith decoder layer calculation and gradient detachment operation [42], respectively. b_{i-1} and b'_{m-1} are the input box information at ith decoder layer and output box information at $m-1$th layer, respectively. Note that the box update in mth layer can be calculated as

$$b_m^{\text{pred}} = \text{Update}\left(b'_{m-1}, \Delta b_m\right) \tag{5}$$

TABLE III
COMPARISON RESULTS WITH DIFFERENT COMPONENTS ON SSDD

Model	CCD	LTTM	AP	AP_{small}
DINO	×	×	65.2	63.2
	√	×	69.7 (+4.5)	66.7 (+3.5)
	×	√	68.8 (+3.6)	68.2 (+5.0)
	√	√	**74.7** (+9.5)	**73.4** (+10.2)

where the term Update(·) indicates the box iterative update process. In our work, we employ the same approach for box update as in deformable DETR [42].

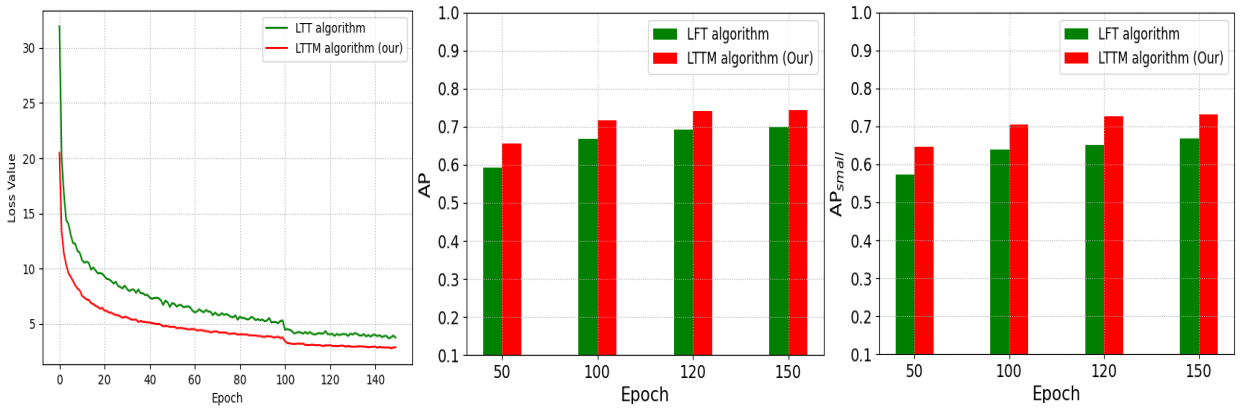
D. Loss Function

As presented in Fig. 1, the total loss of the proposed RDB-DINO consists of two parts: the Hungarian matching and CCD losses, where the matching loss is the same as in DETR [27]. The proposed CCD loss also contains two components: the regression for the box and classification for the category. In this article, we employ the combination of the $L1$ loss [49] and generalized IoU (GIoU) loss [50] as box regression loss L_{Box}, and it can be defined as

$$L_{\text{Box}}(b_i, \widehat{b}_i) = \lambda_{L1} L_1(b_i, \widehat{b}_i) + \lambda_{\text{GIoU}} L_{\text{GIoU}}(b_i, \widehat{b}_i) \tag{6}$$

where $\widehat{b}_i$ and b_i denote the vectors (including center coordinates, height, and width) of the GT and predicted boxes. λ_{L1} and λ_{GIoU} are coefficients to measure the influence of each term. In addition, we use focal loss [14] as the classification

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(a) (b) (c)

Fig. 6. Comparison of different box update algorithms in the training process. (a) Training loss curve. (b) Accuracy on AP at different epochs. (c) Accuracy on AP_{small} at different epochs.

TABLE IV
COMPARISON OF DETECTION RESULTS WITH SOTA DETECTORS ON SSDD

Category	Model	Entire				Inshore		
		AP	$AP_{0.5}$	$AP_{0.75}$	AP_{small}	AP	$AP_{0.5}$	$AP_{0.75}$
Anchor-based	RetinaNet [55]	44.4	77.8	47.8	44.6	24.7	51.8	23.3
	A-BFPN [57]	59.6	96.8	69.9	57.9	–	88.3	–
	GA RCNN [58]	60.3	94.0	69.9	59.9	44.9	79.8	45.7
	Faster RCNN [49]	63.8	91.8	73.5	61.6	44.1	76.8	46.0
	Libra RCNN [59]	64.7	90.2	78.5	63.7	43.7	75.7	46.3
	Cascade RCNN [60]	65.6	92.5	77.8	64.9	51.1	79.0	57.5
	HRSDNet [61]	67.5	92.6	79.8	66.6	52.8	79.9	58.2
	Quad-FPN [62]	–	95.3	–	–	–	84.7	–
Anchor-free	FCOS [13]	57.1	89.1	66.8	57.9	34.7	65.4	32.2
	CP-FCOS [63]	57.3	89.3	65.4	57.5	37.7	71.3	34.8
	Autoassign [64]	59.3	92.2	69.0	60.9	41.2	73.8	39.2
	FEPS-Net [65]	59.9	96.0	67.5	55.1	47.1	85.8	48.9
	ATSS [66]	60.6	91.3	69.4	61.1	40.3	72.4	40.0
	FBUA-Net [32]	61.1	96.2	77.6	59.9	–	–	–
	ATSD [67]	63.7	96.9	74.4	59.8	–	–	–
	GA-Net [34]	64.8	95.7	77.6	64.5	52.5	83.6	57.1
Transformer-based	DETR [27]	55.6	88.3	69.7	54.3	41.2	67.6	45.2
	MGF-Net [68]	–	96.2	–	–	–	85.6	–
	Deformable DETR [45]	58.0	89.9	70.9	57.9	43.3	68.3	47.7
	ESTDNet [69]	59.4	93.8	69.1	55.4	–	–	–
	DINO [29]	65.2	89.4	77.4	63.2	49.4	75.5	52.2
	OEGR-DETR [48]	–	93.9	84.0	–	–	–	–
	CRTransSar [43]	–	97.0	76.2	–	–	–	–
	RDB-DINO (Our)	**74.7**	**98.3**	**90.5**	**73.4**	**68.8**	**96.5**	**83.4**

loss L_{Cls}, and it can be formulated as

$$L_{\text{Cls}}(p_i, \widehat{p}_i) = \begin{cases} -\alpha \cdot \sum_{i=1}^{2N_g} (1-p_i)^{\gamma} \log(p_i) & \widehat{p}_i = 1 \\ -(1-\alpha) \cdot \sum_{i=1}^{N_g} (p_i)^{\gamma} \log(1-p_i) & \widehat{p}_i = 0 \end{cases} \tag{7}$$

where $\widehat{p}_i$ and p_i denote the GT and predicted class probabilities, respectively. N_g signifies the number of CCD groups. The weight coefficient α gauges the relevance between positive and negative queries. γ is a hyper-parameter to balance the impact of classification probability. Therefore, the total loss of the proposed framework can be calculated as

$$L_{\text{Final}} = L_{\text{Box}}(b_i, \widehat{b}_i) + L_{\text{Cls}}(p_i, \widehat{p}_i). \tag{8}$$

According to DETR [27], we apply the auxiliary losses [51] during each decoder training to enhance the capability of feature learning and representation in the network.

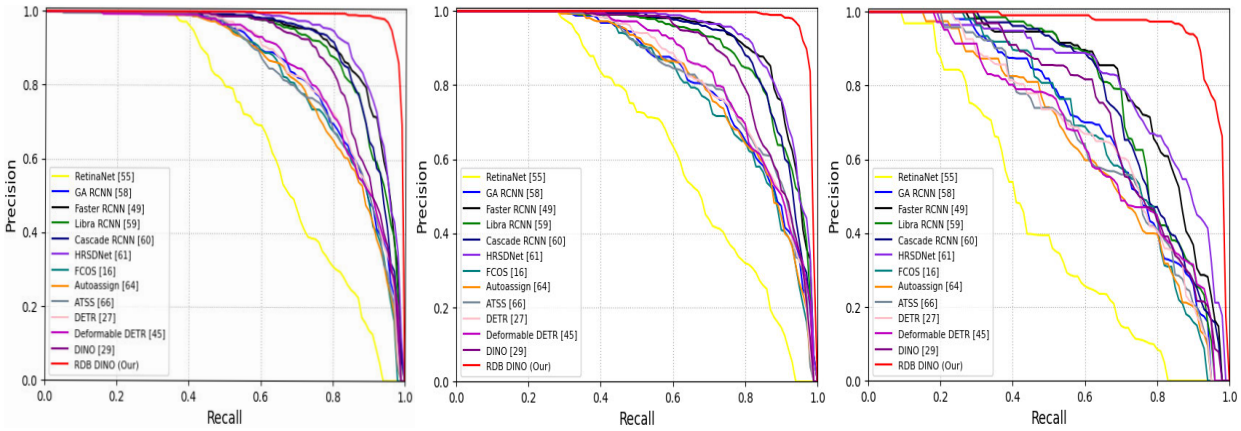
IV. EXPERIMENTS AND DISCUSSION

In this section, we first describe the commonly used public SAR datasets in detail. Next, we introduce the implementation details including parameter selection and evaluation metrics in our experiments. Then, we focus on the investigation and discussion of the proposed approach based on ablation experiments. Finally, extensive experiments are conducted to verify the superiority of the proposed model on open datasets in comparison with existing methods.

A. Datasets

To evaluate the performance of our proposed RDB DINO, the public SSDD [52] and High-Resolution SAR Images

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(a) (b) (c)

Fig. 7. PR curves of different methods on SSDD. (a) PR curves on AP in entire. (b) PR curves on AP_{small} in entire. (c) PR curves on AP in inshore.

TABLE V
COMPARISON OF DETECTION RESULTS WITH SOTA DETECTORS ON HRSID

Category	Model	Entire				Inshore		
		AP	$AP_{0.5}$	$AP_{0.75}$	AP_{small}	AP	$AP_{0.5}$	$AP_{0.75}$
Anchor-based	RetinaNet [55]	56.3	81.8	62.7	57.4	37.0	63.6	37.2
	Faster RCNN [49]	60.0	79.6	63.8	60.5	49.5	68.4	55.8
	Libra RCNN [59]	60.9	80.4	69.9	61.6	49.0	66.7	55.7
	Cascade RCNN [60]	64.0	82.4	72.9	64.4	55.3	72.9	61.6
	HRSDNet [61]	65.2	83.0	74.5	65.6	55.8	73.6	63.1
	Quad-FPN [62]	–	86.1	–	–	–	70.8	–
	GA RCNN [58]	65.5	90.7	74.9	66.7	59.7	87.3	67.3
	FC RCNN [70]	68.6	89.2	77.6	67.4	60.8	–	–
Anchor-free	FCOS [16]	53.7	82.7	59.4	55.1	37.7	69.4	36
	CP-FCOS [63]	55.6	82.9	62.3	56.9	49.3	74.0	52.1
	Autoassign [64]	59.8	86.1	66.7	61.7	46.5	77.1	48.1
	ATSS [66]	60.1	84.6	65.7	61.4	46.0	73.8	47.9
	FEPS-Net [65]	65.7	90.7	74.3	66.8	58.6	84.9	63.3
	GA-Net [34]	66.5	91.1	76.2	68.4	60.9	87.7	69.1
	ATSD [67]	67.3	88.2	–	–	–	–	–
	FBUA-Net [32]	69.1	90.3	79.6	69.6	–	–	–
Transformer-based	PVT-SAR [38]	51.7	87.3	55.6	35.7	–	72.0	–
	DETR [27]	52.9	79.5	62.6	51.3	43.9	64.0	41.1
	Deformable DETR [45]	54.1	81.0	63.5	58.3	47.1	63.2	51.4
	YOLO-SD [40]	59.5	83.7	67.6	62.3	–	–	–
	DINO [29]	63.2	86.7	72.4	64.4	47.6	73.6	52.7
	Trans-ECG [43]	–	87.8	65.5	–	–	–	–
	OEGR-DETR [48]	64.9	90.5	74.4	66.7	–	–	–
	WAFormer [71]	65.1	87.2	75.5	65.8	–	–	–
	RDB-DINO (Our)	**71.8**	**92.8**	**82.4**	**72.5**	**68.8**	**91.2**	**79.5**

Dataset [53] (HRSID) are employed as the touchstones. The main parameters of SSDD and HRSID are shown in Table I.

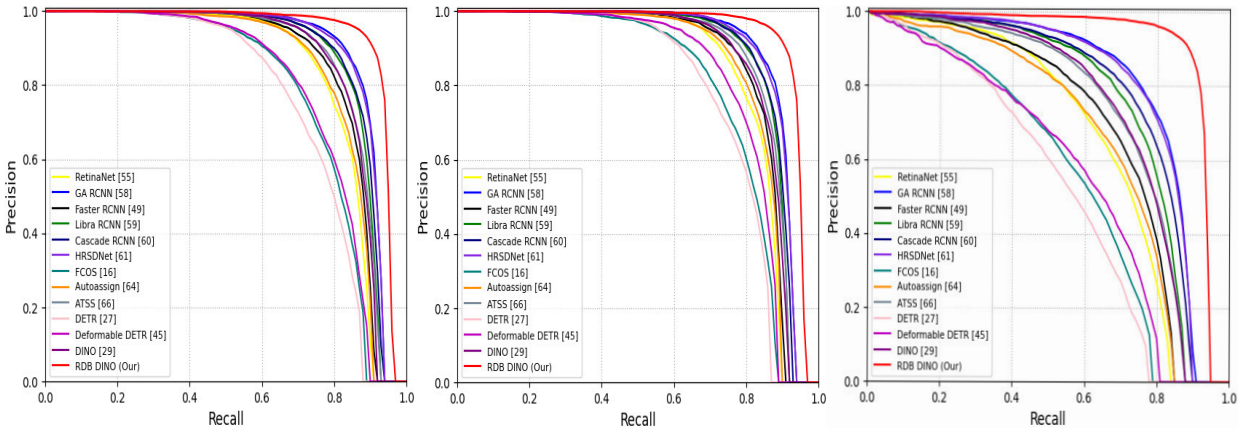
1) *SSDD:* SSDD is the first open dataset for SAR ship detection, there are 1160 images with 2456 ships in total, which are categorized into training, test, and validation sets in a ratio of 7:2:1. In SSDD, the test set contains 46 images, while the number of small ships is 1529, accounting for over 60% of the total.
2) *HRSID:* HRSID is a large and high-resolution ship dataset with 5604 SAR images in 800 × 800 pixel-size. There are 16 965 ships in total. In HRSID, all the images are divided into the training set with the amount of 65% images, and the test set with 35% images. There are 9242 small-scale ships in the dataset.

It is can be observed from Table I that the SSDD mainly contains small ships occupied less than 32 × 32 pixels, while the HRSID has a large amount of ship targets.

B. Implementation Details and Evaluation Metrics

In this article, the ResNet-101 [25] network pretrained on ImageNet is regarded as the backbone with the initial learning rate of 0.0001 in training. Meanwhile, we adopted the Adam weight decay regularization (AdamW) [54] optimization algorithm with a weight decay of 0.0001 at 100th epoch. Considering the different amounts of data for the two

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(a) (b) (c)

Fig. 8. PR curves of different methods on SSDD. (a) PR curves on AP in entire. (b) PR curves on AP_{small} in entire. (c) PR curves on AP in inshore.

TABLE VI

MODEL SIZE AND INFERENCE SPEED COMPARISONS WITH TYPICAL METHODS

Categroy	Model	Backbone	Parameter (M)	FPS (HRSID)	FPS (SSDD)
CNN-based	RetinaNet [55]	ResNet-50	36.1	22.5	25.8
	A-BFPN [57]	ResNet-101	60.5	–	7.5
	Faster RCNN [49]	ResNet-50	43.2	24.9	28.9
	Cascade RCNN [60]	ResNet-101	68.9	21.2	24.1
	Libra RCNN [59]	ResNet-50	41.7	24.7	28.6
	GA RCNN [58]	ResNet-50	41.7	20.2	22.8
	HRSDNet [61]	ResNet-101	77.2	7.4	7.7
	Quad-FPN [62]	ResNet-50	–	13.6	11.4
	FC RCNN [70]	ResNet-101	–	–	–
	FCOS [16]	ResNet-50	**31.8**	**26.5**	29.7
	CP-FCOS [63]	ResNet-50	36.0	23.1	26.0
	Autoassign [64]	ResNet-50	36.0	22.7	26.1
	ATSS [66]	ResNet-50	31.9	22.7	26.2
	FEPS-Net [65]	ResNet-50	37.3	10.6	31.5
	FBUA-Net [32]	ResNet-50	36.5	12.2	**31.8**
	ATSD [67]	DLA-34	61.5	–	31.3
	GA-Net [34]	ResNet-50	43.4	16.4	20.0
Transformer-based	PVT-SAR [38]	Swin transformer	121.0	6.8	–
	DETR [27]	ResNet-101	50.6	17.3	20.0
	ESTDNet [69]	Swin transformer	51.5	–	13.3
	Deformable DETR [45]	ResNet-101	50.2	18.2	20.6
	YOLO-SD [40]	CSPDarkNet53	59.6	8.6	–
	DINO [29]	ResNet-101	59.2	18.8	17.5
	WAFormer [71]	Swin transformer	41.3	–	–
	CRTransSar [42]	Swin transformer	96.0	–	7.5
	Trans-ECG [43]	Swin transformer	33.6	13.0	–
	RDB-DINO (Our)	ResNet-101	64.2	18.3	16.8

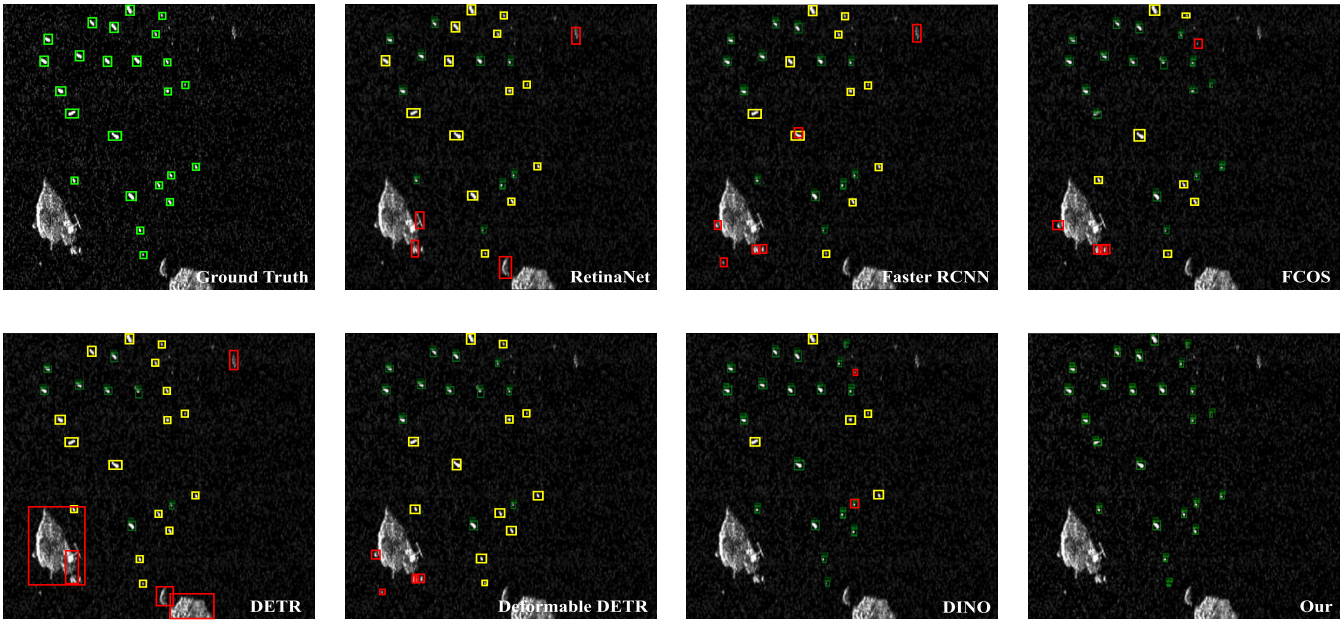
TABLE VII

TRAINING EFFICIENCY FOR DIFFERENT TRANSFORMER-BASED DETECTORS WITH RESNET-101

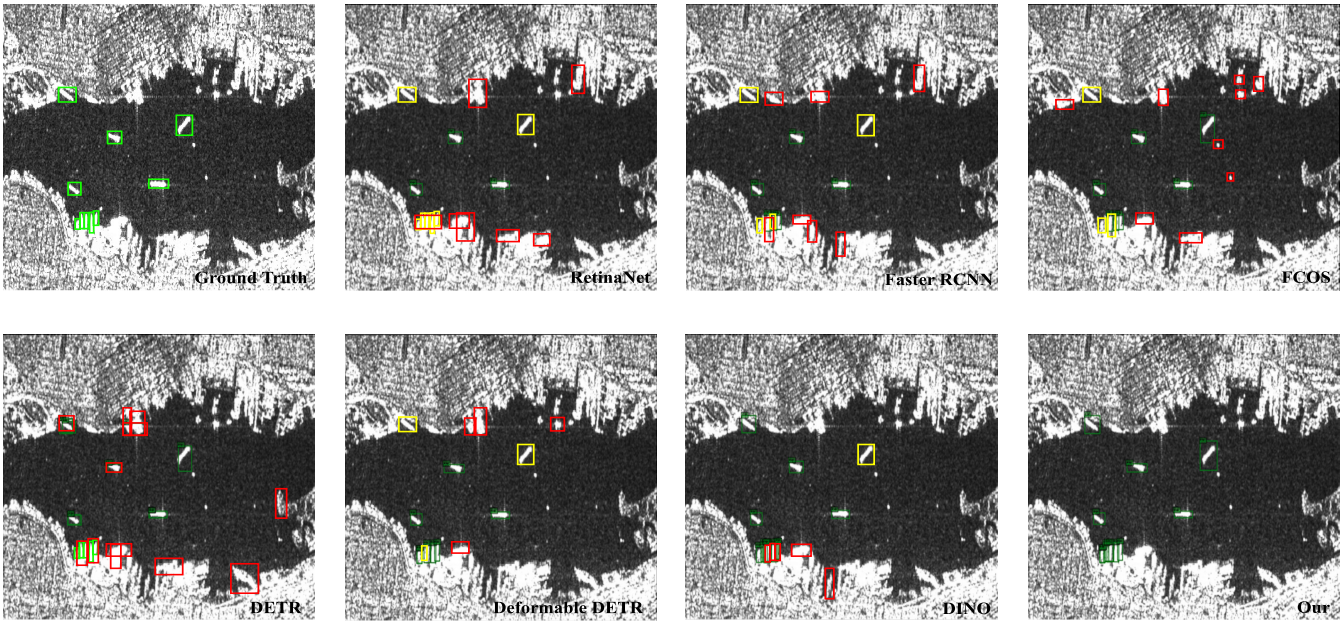
Model	Batch Size	GPU Memory	Training Time	Epoch	AP
DETR	4	67GB	~10 min/epoch	300	52.9
Deformable DETR	3	44GB	~9 min/epoch	150	54.1
DINO	3	44GB	~7 min/epoch	150	63.2
RDB-DINO (Our)	3	44GB	~7 min/epoch	35	**64.0**

datasets (the number of images in HRSID is significantly more than that in SSDD), we set the batch size to 2 and 3 for SSDD and HRSID, respectively. Furthermore, the training epoch number is selected as 150. Our experiments are implemented under Pytorch 1.9.0, CUDA 11.1, and performed on an NVIDIA Tesla A100 GPU card with 80-GB memory. The specific parameter setting is illustrated in detail in Table II.

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(a)



(b)

Fig. 9. Several visualization instances of existing methods in different scenes on SSDD. The green, red, and yellow boxes denote the GTs, false alarms, and missed detection, respectively. (a) Ship detection in several small ships distribution scenarios with minor interference. (b) Ship detection in complex inshore scenarios with few small ship targets.

In addition, six measurements are employed to assess the performance of the proposed model quantitatively, including the average precision (AP), AP_{50}, AP_{75}, AP_{small}, Parameter, and frames per second (FPS). AP indicates the AP of 10 values ranging from the IoU thresholds of 0.5 to 0.9, with an interval of 0.05. The AP can be defined as

$$\mathrm{AP} = \int_0^1 P(R)dR \tag{9}$$

where P and R indicate the recall rate and precision rate, respectively, and they can be calculated as

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

where TP, FP, and FN denote true positive, false positive, and false negative, respectively. Generally, AP is used to evaluate the overall detection and robustness performance [55]. AP_{50} and AP_{75} are the detection accuracy when the IoU thresholds are equal to 0.5 and 0.75, respectively. AP_{small} represents detection accuracy for small ships with labels less than 32×32 pixels, and it focuses on the ability of a detector in small object detection [56]. The parameter is applied to estimate the model size. FPS means the inference speed of the network.

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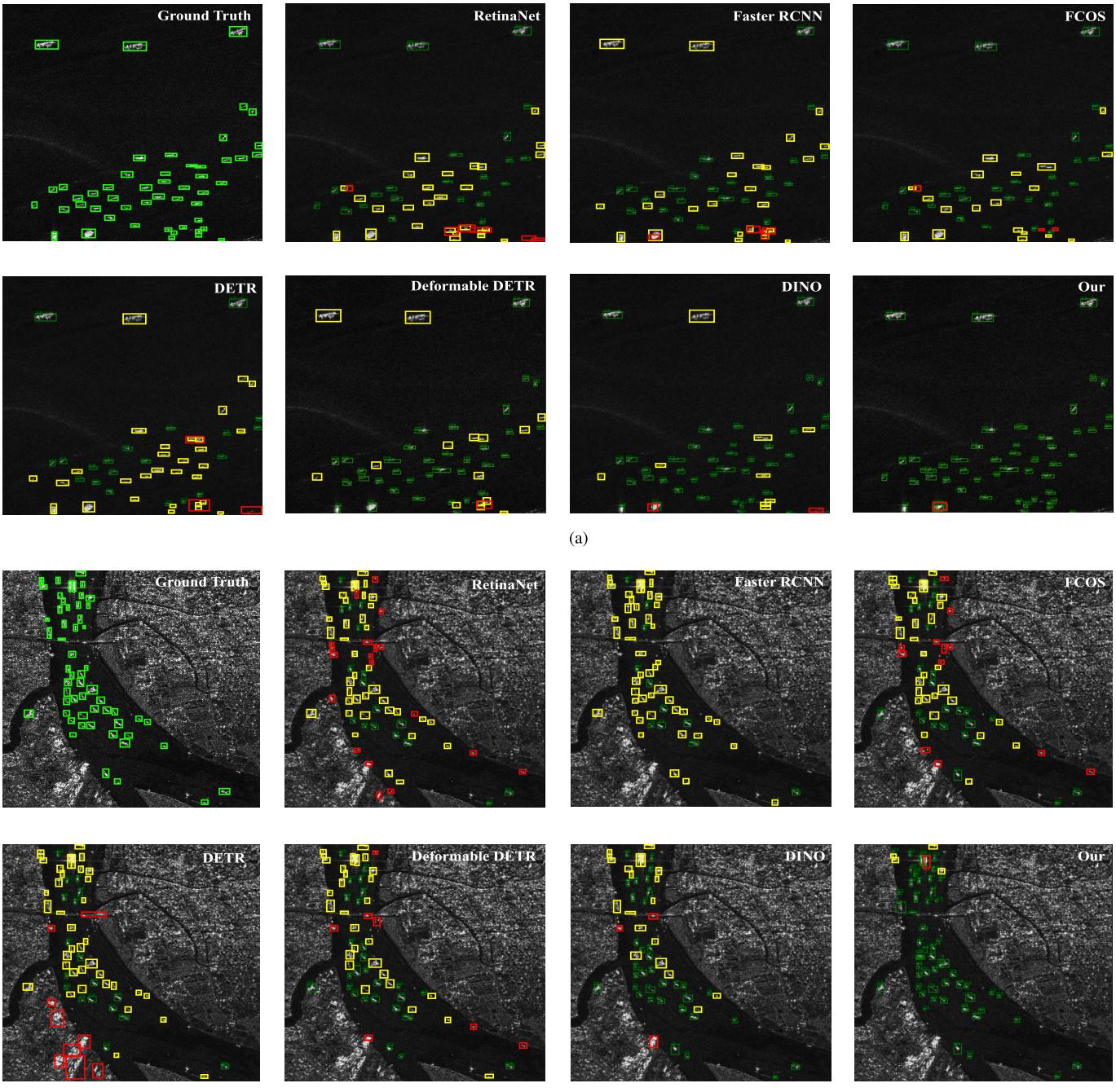


Fig. 10. Several visualization instances of existing methods in different scenes on HRSID. The green, red, and yellow boxes denote the GTs, false alarms, and missed detection, respectively. (a) Ship detection in plentiful small ships dense distribution scenarios. (b) Ship detection in complex inshore environments with large numbers of small ship targets.

C. Ablation Study

To validate the effectiveness of our proposed components, a series of ablation experiments are conducted on SSDD under the same parameter setting for fair comparison. Two modifications are designed to facilitate the baseline more effective in the representation of ship targets, especially for small-scale ships. The overall quantitative comparisons based on the DINO [29] are presented in Table III. As we can observe, both proposed techniques offer significant improvements over the baseline network in terms of accuracy. Compared with the baseline, the proposed CCD strategy achieves 4.5% and 3.5% higher on AP and AP_{small}, respectively. Meanwhile, our LTTM algorithm increases the AP and AP_{small} by 3.6% and 5.0%, respectively. Finally, AP and AP_{small} gain the major enhancements of 9.5% and 10.2%, respectively, by integrating both CCD and LTTM. In the following parts, the impact of each component in the proposed model will be elaborated and analyzed in detail, separately.

1) Effect of CCD: We first investigate the influence of the different training strategies. As presented in Fig. 5, we select a complex inshore scenario with densely parked ships for detection visualization under different training techniques.

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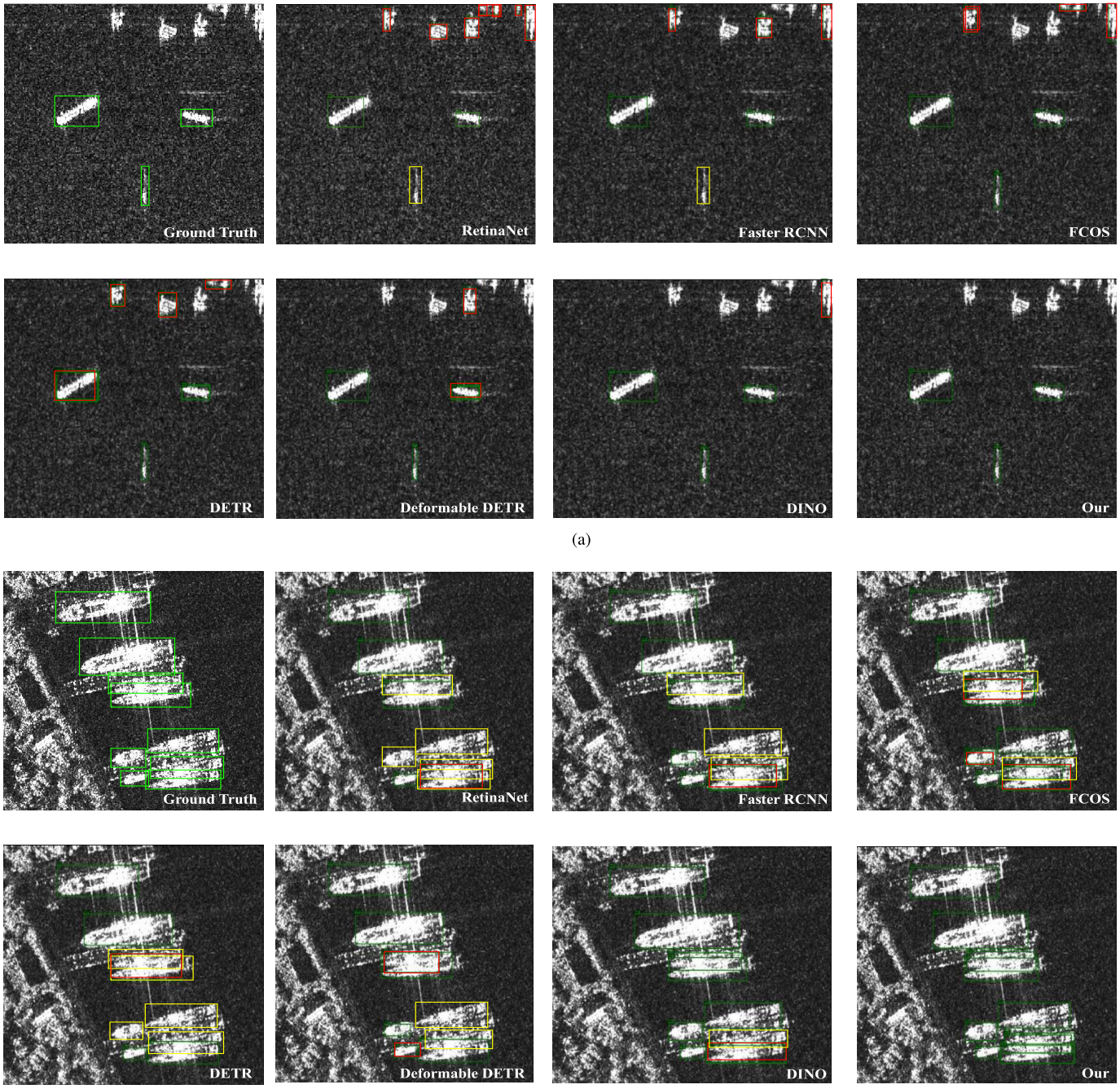
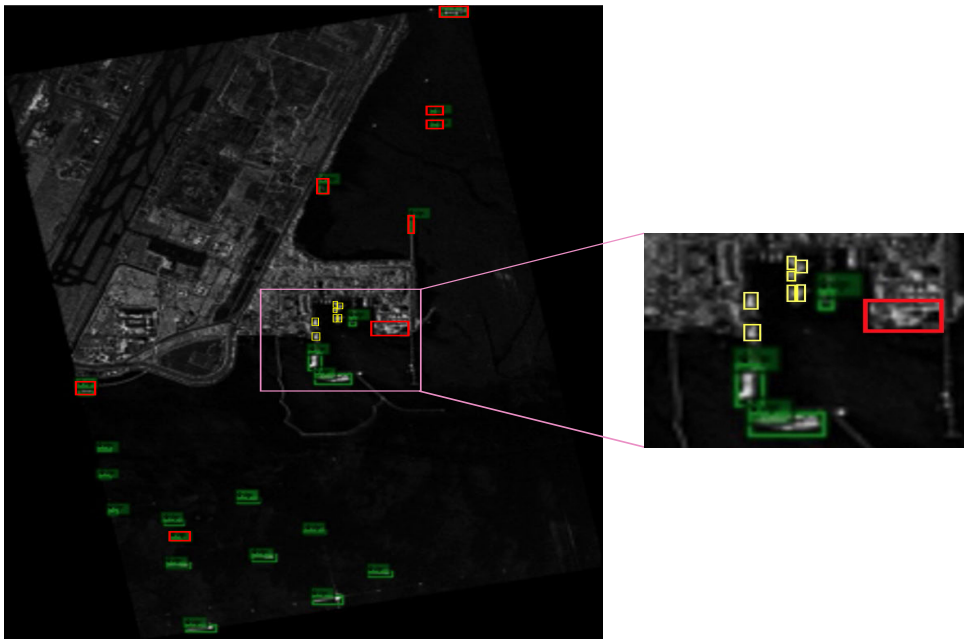


Fig. 11. Several visualization instances of existing methods in different scenarios under complex sea conditions. The green, red, and yellow boxes denote the GTs, false alarms, and missed detection, respectively. (a) Ship detection in offshore scenarios. (b) Ship detection in inshore scenarios.

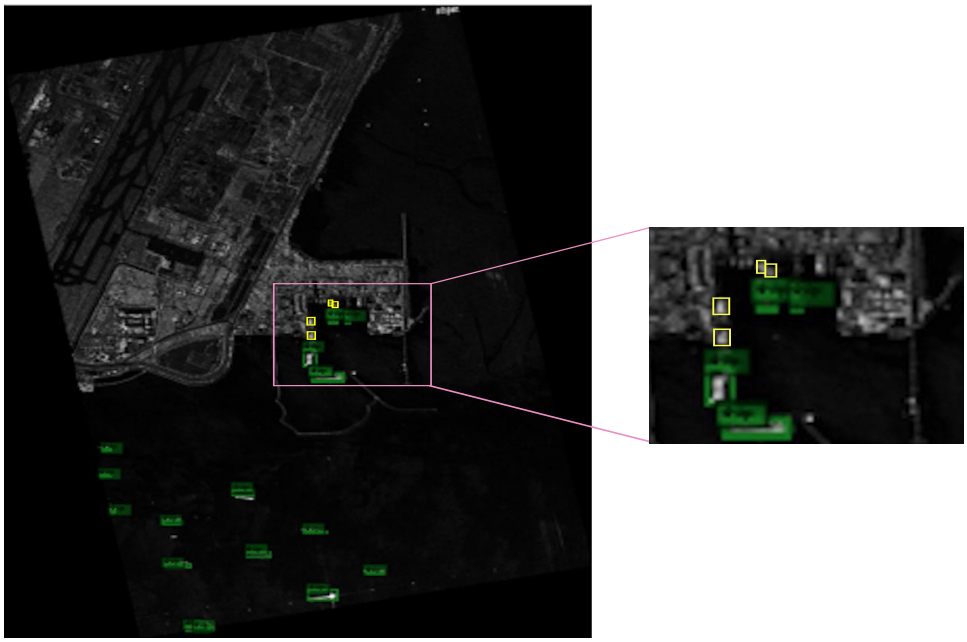
As we can observe from Fig. 5(b) that the model trained using CDN [29] has a decent effect on dense ships with significant differences, but has a mediocre performance for dense ships with similar shapes. This is because the CDN [29] inputs positive and negative queries with hard labels via introducing different noise ratios to the network, enabling the network to distinguish ship targets with large differences in complex environments. However, limited by the rough hard label, the CDN [29] technique lacks an accurate representation of embedding for positive samples, resulting in insensitivity detection to ship targets with high similarity. In contrast, the proposed RDB DINO with CCD training strategy can overcome the interference between similar targets to accurately locate ships, as shown in Fig. 5(c). This is because our CCD aggrandizes the sample information by adding GT queries, and modifies the label embeddings of queries based on IoU, which not only enables the network to fully learn the information from different samples, but also promotes the network to have better discrimination for samples with high similarity.

2) Effect of LTTM: The training loss curves with different box update algorithms are reported in Fig. 6(a). Intuitively, compared with the model guided by LFT, the proposed LTTM-guided model yields excellent performance with a lower loss value throughout the training. Meanwhile, our

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(a)



(b)

Fig. 12. Detection visualization instance in larger-scale SAR image. The green, red, and yellow boxes denote the GTs, false alarms, and missed detection, respectively. (a) Detection results of DINO. (b) Detection results of the proposed RDB-DINO.

model exhibits consistently high accuracy on both AP and AP_{small} at different training epochs, as shown in Fig. 6(b) and 6(c). In summary, the proposed LTTM can assist the network to achieve stable training and satisfactory detection accuracy.

D. Detection Performance Comparisons

To demonstrate the superior performance of our detector, we compare the proposed RDB-DINO with the existing SOTA detectors on the SSDD and HRSID. First, we conduct experimental comparisons in terms of accuracy to show the feasibility of our method on two datasets. Next, we illustrate the visualization of detection results under several typical scenarios to present the effectiveness of our method. Finally, we compare the model parameter size and inference speed of existing methods. The detection results on SSDD and HRSID are listed in Tables IV and V. Intuitively, our RDB-DINO surpasses existing detectors in all metrics on both datasets. Moreover, the proposed approach demonstrates more significant performance benefits in complex inshore scenarios. Specifically, compared with the current SOTA results, the proposed model achieves 7.2%, 6.8%, and 16.0% higher on AP, AP_{small} of entire, and AP of inshore, respectively, on SSDD, while outperforms by 2.7%, 2.9%, and 7.9% on AP,

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AP_{small} of entire, and AP of inshore, respectively, on HRSID. In addition, the corresponding PR curves on two datasets are drawn in Figs. 7 and 8. It can be observed that the proposed network obtains better results with the maximum enclosed area compared to other detectors. These result comparisons manifest that the proposed RDB-DINO has superior small-ship detection performance, especially in complex environments. Furthermore, we select several typical scenarios on SSDD and HRSID datasets to visualize detection results for an intuitive comparison of existing methods' performance, as presented in Figs. 9 and 10. Fig. 9(a) and (b) presents the detection results in several small ships distribution scenarios with minor interference, and the complicated environments with few small ship targets, respectively. It can be seen that the existing detectors suffer from missed detections and false alarms, while the proposed network can accurately detect each ship target. Fig. 10(a) and (b) indicates the detection results in plentiful small ships dense distribution scenarios and complex inshore environments with large numbers of small ships, respectively. As we can observe that our RDB-DINO outperforms other competitors with fewer errors and better accuracy. To demonstrate the technical superiority of the proposed method, we visualize the comparative detection results of existing methods in different scenarios under complex conditions where ships and sea areas are hardly distinguished, as shown in Fig. 11. Intuitively, the proposed model can suppress the interference to exhibit improved detection accuracy both in offshore and inshore scenes. Fig. 12 illustrates the detection results within a larger-scale SAR image. It can be observed that the proposed method has superior performance and outcomes with more correct detections and fewer errors compared with the baseline, especially for small-scale ship detection in inshore scenarios. Regarding the model size and test speed comparisons as shown in Table VI, the CNN-based detectors generally have fewer parameters and faster speed than those of transformer-based detectors. In addition, the proposed model can obtain a feasible computation-efficient with tolerable computation complexity, compared with existing transformer-based detectors.

E. Training Efficiency Discussion

Due to the global modeling capability and massive computational complexity of transformer structure, transformer-based detectors generally exhibit larger sizes compared to CNN-based methods, as evidenced in Table VI. Therefore, the training efficiency should be a concern. Considering the characteristics of the dataset, we choose HRSID with large amounts of data as the benchmark to assess the training efficiency of our model. The GPU memory and training time for a typical E2E transformer model is reported in Table VII. It is evident that our model utilizes satisfactory GPU memory in training and achieves superior results in the shortest training time. In conclusion, our model exhibits efficiency for training.

V. Conclusion

This work proposes a novel E2E transformer-based framework for small-scale ship detection in SAR images, A CCN training strategy is developed to exploit different query information in transformer decoder components. In addition, we design an LTTM algorithm for box refinement to boost the target update information, which is beneficial for improving the detection accuracy of small ships. Substantial comparative experiments on commonly used public datasets manifest the superiority of our network in terms of detection performance and training efficiency. In the future, we will investigate a more efficient E2E transformer-based ship detector to accelerate inference speed without remarkable accuracy degradation, while also addressing the issue of sidelobe interference.

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